Part 1: Concept White Paper

Basic Information	
Company Name: McLean Capital Development LLC	
Proposal Title: Independent Evaluation of Nanoscale Energy Conversion Claims for Army Sustainment	
TRL: 2 (self-assessed; concept stage)	**Location:** Newport Beach, CA, USA (mailing)
Number of Employees: 4	**Previous DoW Business:** No
Capital Raised Total: Approx. USD 10,000 (including founder contributions)	**Website:** https://energywave.ai
Primary Contact Name: Ian Franklin McLean	**Primary Contact Email:** ian@mcleancompanies.com
Secondary Contact Name: Chip Story	**Secondary Contact Email:** crstory@sbcglobal.net

Part 1: Introduction (5%)

EnergyWave, a project of McLean Capital Development LLC, proposes an independent measurement platform to evaluate nanoscale energy-conversion claims before they drive Army power-system investment. The first candidate is a published metal-insulator-metal (MIM) diode and Casimir-cavity experiment, tested alongside matched controls under predeclared criteria. Guarded electrical measurements, artifact tests and thermal bounds would test whether a signal survives specified controls; source attribution would depend on quantified bounds. This is a concept-stage evaluation capability, not a power-source offering. EnergyWave has no verified prototype data; a null result is an expected, useful outcome. Independent laboratory execution and a qualified technical lead are planned but not secured.

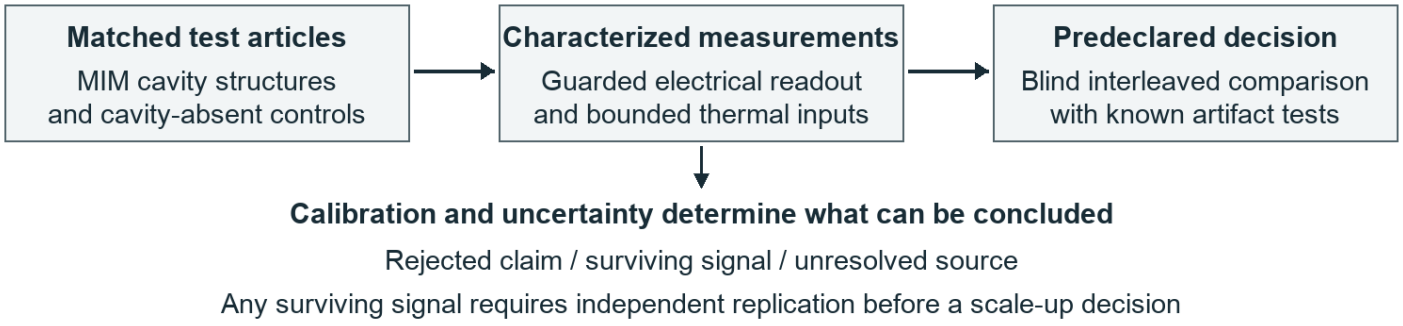


Figure 1. Proposed evaluation path for Army sustainment decisions. The platform would compare MIM cavity structures with matched controls and account for conventional electrical and thermal effects. Full energy-source attribution depends on achieved uncertainty. This is a conceptual protocol, not built hardware, measured results or a committed laboratory partnership.

Part 2: Army Benefits (25%)

Solution's Advantages. For Adaptive Sustainment, the intended advantage is an auditable decision record before investing in uncertain power-source claims. Alternatives include literature review and conventional low-current testing. This proposal combines matched fabricated controls, blinded measurements and explicit energy-accounting limits. Superior accuracy, speed or cost remains unproven. Established batteries, generators and converters remain power options; the proposed instrument does not replace them.

Solution's Impact. A credible rejection could prevent investment in a signal caused by known effects. A signal surviving the protocol could justify independent replication, not immediate scale-up. Potential Army value is better evidence for selecting sustainment technologies. No endurance, mass, resupply or power-density improvement is claimed. Those operational benefits would require a validated energy source, an engineered system and comparison under an Army-relevant duty cycle.

Degree of Innovation. The proposed contribution combines cavity-specific controls, low-current measurements, thermal bounds and a predeclared decision process. Electrometry, nanofabrication and calorimetry are established methods. The platform's differentiation and ability to yield more decisive evidence remain to be demonstrated; no first-of-kind capability or new physical energy source is claimed.

Part 3: Technical Approach (40%)

Scientific Feasibility. Under the second law, a passive device at thermal equilibrium cannot sustain net electrical output without an energy input. Ordinary thermal/radiative inputs, stored energy, leakage, contact effects and instrument bias must be considered first. All inputs must balance output, losses and stored-energy changes. Measure chip and enclosure temperatures and impose controlled gradients; responses may reflect conventional conversion, not automatically artifacts. Electrical sensitivity alone cannot establish energy balance. If achieved thermal uncertainty cannot exclude conventional energy paths, source attribution remains unresolved. The experiment tests the published claim without assuming its mechanism.

Enabling Technologies. Commercial electrometry and guarded wiring are established; instrument access and chain calibration remain unverified. Matched MIM/control fabrication is unbuilt with no committed facility; process qualification and materials characterization must establish geometry and comparability before testing. Temperature-controlled stages are established, but integrated thermal source attribution is exploratory: its go/no-go gate is achieved sensitivity to relevant energy paths. Calibration must bound noise, drift and leakage. Sample counts and acceptance criteria require technical review. Individual numerical TRLs remain unassessed; component maturity does not establish system maturity.

Technical Team. Ian McLean coordinates the program. The technical investigator must have demonstrated low-signal instrumentation and thermal-metrology experience. Independent laboratory execution and skeptical external protocol review are planned; no laboratory, investigator or institutional partnership is committed in this proposal. Relevant personnel, credentials, effort and access must be confirmed before execution. AI-assisted document critique is not experimental validation.

Data Quality, Technical. Moddel et al. reported 1.4 pW for a 33 nm cavity [1]; EnergyWave has not replicated this. Maclay analyzed the proposed mechanism [2]. Subject to qualified-lead review and fabrication feasibility, predeclare expected current polarity, cavity-thickness effects and reversed-lead responses, accounting for optical/thermal confounds. Evidence would include defined-load measurements, calibration, uncertainty bounds, matched controls and blinded interleaved runs. No exact geometry or sensitivity is promised. Outcomes may reject a claim, retain a signal or remain inconclusive. Null findings constrain tested devices and conditions; fabrication differences must be assessed before interpreting non-reproduction.

Part 4: Commercial Potential (25%)

R&D to Product Revenue. Potential customers are organizations evaluating low-power conversion devices, including laboratories, technology developers and research agencies. Initial independent testing would be a service; service receipts and grants are not product revenue. A possible product is a characterized bench and reproducible protocol for laboratories, contingent on demonstrated performance, repeatability and an evidenced customer need. The path is to validate the measurement capability, obtain customer requirements, test an engineering package with a pilot user and establish a viable sales model. No product revenue, letter of intent, pilot agreement or signed customer contract is claimed.

Competitive Edge. Any durable advantage would need to come from proven integration, reusable calibration and control methods, relevant data and specialized execution capability. These are development objectives, not an established moat. Existing electrometers and laboratory services are credible alternatives; no claim that competitors or suitable instruments do not exist is made. Current patent status, ownership and disclosure rights require counsel verification. EnergyWave does not claim here that company-owned or protected IP assures an advantage.

Other People's Money. No committed non-Army development funding is established for this proposal. Private investment, commercial co-development and suitable research programs are possible future sources, but not an assured runway. Any prior, current or pending federal support must be checked for overlap and disclosed as required. A later Army SBIR/STTR application, if invited, is a separate process. The present entry does not assume a prize, follow-on award or product sale.

Development Gates. A possible six-month feasibility sequence, subject to people, access and quotes, is: months 1-2, select a qualified independent lead through documented technical review and secure fabrication/characterization access, or defer execution; agree specimen design and electrical/thermal uncertainty limits; months 3-4, obtain matched test articles and qualify the chain; months 5-6, run a bounded blinded comparison if calibration gates pass. If fabrication or thermal sensitivity is inadequate, report the limitation rather than promise full replication. Independent reproduction and product development are later gated work. Deliverables would be the protocol, raw-data index, uncertainty budget and a justified rejection, surviving-signal or inconclusive finding.

GovThink is a proposed governance initiative intended to support documented safety assessments, independent technical review and controlled sharing of appropriately cleared research. For civilian humanoid-robotics applications, proposed requirements would address electromagnetic compatibility, tamper resistance and authorized-system use, subject to engineering validation. Any defense-related work would require separate legal, export-control and security review before disclosure or development. These are proposed safeguards, not verified capabilities, certifications or claims of government affiliation. Appropriate government safety and regulatory experts are warmly invited to discuss potential collaboration, without implying endorsement, partnership or commitments.

References. [1] Moddel et al., Optical-Cavity-Induced Current, Symmetry 13, 517 (2021), doi:10.3390/sym13030517. [2] G. J. Maclay, Modeling Electronic Devices with a Casimir Cavity, Physics 6(3), 1124-1131 (2024), doi:10.3390/physics6030070.